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Shrimp microbiome and immune development in the early life stages

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ABSTRACT

With its contribution to nutrition, development, and disease resistance, gut microbiome has been recognized as a crucial component of the animal's health and well-being. Microbiome in the gastrointestinal tract constantly interacts with the host animal's immune systems as part of the normal function of the intestines. Interactions between the microbiome and the immune system are complex and dynamic, with the microbiome shaping immune development and function. In contrast, the immune system modulates the composition and activity of the microbiome. In shrimp, as with all other aquatic animals, the interaction between the microbiome and the animals occurs at the early developmental stages. This early interaction is likely essential to the development of immune responses of the animal as well as many key physiological developments that further contribute to the health of shrimp. This review provides background knowledge on the early developmental stage of shrimp and its microbiome, examines the interaction between the microbiome and the immune system in the early life stage of shrimp, and discusses potential pitfalls and challenges associated with microbiome research. Understanding the interaction between the microbiome and shrimp immune system at this crucial developmental stage could have the potential to aid in the establishment of a healthy microbiome, improve shrimp survival, and provide ways to shape the microbiome with feed supplements or other strategies.

1. Introduction

A microbiome is the complex community of microorganisms, including bacteria, fungi, and viruses, that inhabit a particular environment, such as the gastrointestinal tract of animals. The gut microbiome plays a critical role in host health by contributing to nutrient absorption, immune system development, and protection against pathogens (Visconti et al., 2019). In shrimp, the microbiome is particularly important during early development because it can influence the development of the shrimp's immune system (Angthong et al., 2020, 2021). The early stages of a shrimp's life are characterized by rapid growth and development. In particular, newly hatched shrimp have a sterile, immature digestive system before they mature into nauplii, zoea, mysis, and postlarvae (Motoh, 1985). After the nauplius stage, shrimp such as the black tiger shrimp, *Penaeus monodon*, begin to feed on microalgae or live feeds, allowing bacteria from external sources to colonize in the host gut (Jiravanichpaisal et al., 2007). The interactions between host immunity and non-pathogenic bacteria during the early life stages are critical for the further development of the host immune system (Arrieta et al., 2014; Bates et al., 2006; Tamburini et al., 2016). A healthy microbiome can provide the shrimp with essential nutrients, promote the development of its digestive system, and protect it from harmful microorganisms (Holt et al., 2021). An imbalanced microbial population in early life could make them vulnerable to environmental stressors and pathogens.

This review summarizes the current understanding of the importance of the interaction between the microbiome and the immune system in the early life stage of shrimp, which may have significant implications for aquaculture. The factors that influence the gut microbiota of shrimp and the evidence for the interplay between the host immune system and its gut microbiota are important first steps in understanding the factors that shape the gut microbiome at this critical stage. This knowledge can be used to identify specific bacterial species or communities that are important for the development and maintenance of a healthy gut microbiome. The early life stages of shrimp are a critical period for the establishment of a healthy gut microbiome, and the use of feed supplements such as probiotics during this time can effectively promote the growth and survival of beneficial bacteria in the gut. In addition,

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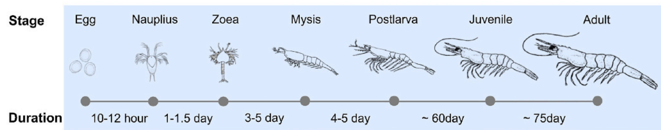

Fig. 1. The life cycle of penaeid shrimp includes egg, nauplius, zoea, mysis, postlarva, juvenile, and adult.

research on the shrimp gut microbiome has evolved in recent years, and there is still much to learn about its complex interactions. New and innovative approaches, such as next-generation sequencing platforms, will allow us to gain new insights and generate more data as the field grows. This review also includes challenges and pitfalls that researchers must consider in order to gain meaningful and accurate data interpretation.

2. Microbiome in early developmental stages of shrimp

2.1. Developmental stage of shrimp

Being crustaceans, shrimp undergo periodic cyclic molting to shed and renew their exoskeletons, which is essential for shrimp growth, metamorphosis, and reproduction (Ronquillo et al., 2006). Shrimp develop through a series of eggs, nauplii, zoea, mysis, postlarvae, juveniles, and adults (Fig. 1). After hatching from eggs, they metamorphose into six nauplius stages, three zoea stages, three mysis stages and postlarval stages before becoming juvenile and adult shrimp (Motoh, 1985; Ronquillo et al., 2006). Their morphology, physiology, ecology and feed requirements change drastically during the early life stages. When shrimp hatch at the nauplius stage, they feed on their reserves, and their digestive systems are still developing before they reach the zoea stage. Zoea shrimp begin feeding on phytoplankton such as *Thalassiosira* sp. and *Chaetoceros* sp. until they reach mysis. Mysis shrimp can feed on phytoplankton and zooplankton such as rotifers or brine shrimp (*Artemia*). Once they develop to the postlarval stage, they will be fed with *Artemia* in combination with artificial feed pellets (Jiravanichpaisal et al., 2007).

2.2. Evidence of factors that influence the development of shrimp microbiome

The gut microbiota is known for their important roles in host metabolism, physiology, and immune system (Bikel et al., 2015; Ringø et al., 2016; Round and Mazmanian, 2009). Several factors influence the composition of the gut microbiota, including developmental stages, nutritional diets, rearing environment and disease (Angthong et al., 2020; Chaiyapechara et al., 2022; Niu et al., 2019; Wang et al., 2020a). The bacterial communities that may colonize the gut of animals after hatching or birth could have a long-term impact on the health status of the host. Therefore, an understanding of the intestinal microbiota since the early life stages of shrimp is necessary. Generally, major bacterial phyla associated with shrimp at early life stages consist of Pseudomonadota (Proteobacteria), Bacteroidota (Bacteroidetes), Planctomycetota (Planctomycetes), Actinomycetota (Actinobacteria), and Bacillota (Firmicutes) (Angthong et al., 2020; Valle-Gough et al., 2022; Vinay et al., 2022; Wang et al., 2020b; Zeng et al., 2017). Several studies have reported that bacterial composition associated with shrimp was different across all developmental stages. For example, the most abundant bacterial phylum in the nauplius stage of black tiger shrimp (*Penaeus monodon*) is Pseudomonadota, such as *Pseudoruegeria*, *Thalassotalea*, *Vibrio,* and *Pseudoalteromonas*, while Planctomycetota (genera *Nautella,* and *Bastopirellula*) and Bacteroidota (genus *Aureispira*) were abundant in zoea (Angthong et al., 2020). Bacterial communities of mysis were shifted to the phylum Bacteroidota including *Aquimarina*, *Spongiimonas*, *Tenacibaculum* and *Spongiimonas*. Once the shrimp reach postlarval stage, the dominant bacteria are Pseudomonadota such as *Thalassobius*, *Vibrio,* and *Loktanella* (Angthong et al., 2020). In Pacific white shrimp (*Litopenaeus vannamei*) at the nauplius stage, the dominant phyla were Pseudomonadota (genera *Vibrio*, *Reinekea* and *Pseudoalteromonas*), Bacteroidota (genera *Pseudofulvibacter*, *Nonlabens* and *Crocinitomix*) and Actinomycetota (genera *Pseudonocardia* and *Gordonia*). At the zoea stage, bacterial communities shifted to Bacteroidota (genus *Algoriphagus*) and Pseudomonadota (genera *Roseovarius* and *Donghicola*), while the relative abundance of Planctomycetota (genus *Rhodopirelluta*) and Actinomycetota (genus *Ilumatobacter*) increased at mysis shrimp. The bacterial communities of the postlarval *L. vannamei* shrimp are shifted to phylum Bacillota (genera *Lactobacillus* and *Bacillus*) and Pseudomonadota (genera *Ruegeria* and *Pseudomonas*) (Wang et al., 2020b). Moreover, bacterial genus *Vibrio* is commonly distributed in the marine environment and has been reported to be associated with high relative abundance in aquatic animals, including shrimp (de Souza Valente and Wan, 2021; Rungrassamee et al., 2013). Previous studies reported that *Vibrio* species established since the early life stages of *P. monodon* (Angthong et al., 2020; Rungrassamee et al., 2013) and *L. vannamei* (Wang et al., 2020b; Zheng et al., 2016) as a natural microflora that resides in host shrimp. Some *Vibrio* strains might become pathogenic bacteria in shrimp under unsuitable conditions such as *V. harveyi* and *V. parahaemolyticus* (Chen et al., 2017b; Rungrassamee et al., 2016), while some strains have been reported for probiotics that increase disease resistance such as *V. alginolyticus*, *V. diabolicus* and *V. hepatarius*, which against acute hepatopancreatic necrosis disease (AHPND)-causing *V. parahaemolyticus* (Paopradit et al., 2021; Ramirez et al., 2022).

It has been known that the changes in physiological and morphological development during the early life stages could influence their microbiota composition. Bacterial communities associated with the postlarva stage of *P. monodon* (5- and 15-day old postlarvae) (Angthong et al., 2020) and Indian white shrimp (*P. indicus*; 6- and 12-day old postlarvae) (Vinay et al., 2022) were similar, those from the earlier stages of egg, nauplius, zoea, and mysis were different, suggesting that the morphological and physiological properties of the developed shrimp digestive system might be responsible for the host-driven selection of the microbiome composition from the postlarval stage. Moreover, the shifts in host nutritional intake and diet requirement at each stage can lead to differences in larval bacterial communities. During shrimp larval development, the nauplius stage fed on their reserves (Jiravanichpaisal et al., 2007), the bacterial community could mainly originate from their parents and hatching environments (Pangastuti et al., 2010; Wang et al., 2020b). When the larvae develop into zoea stage, where they open their mouth and start eating phytoplankton (Jiravanichpaisal et al., 2007; Wang et al., 2020b), their intestinal microbiota begin to establish. Previous studies have shown that the composition of the gut microbiota during the developmental stages of shrimp is related to the type of feeding, which is responsible for the change in bacterial composition. *Tenacibaculum,* belonging to the phylum Bacteroidota was the most abundant genus found in zoea and mysis stages of *P. monodon* (Angthong et al., 2020), and it has been reported to be associated with the marine phytoplankton *Thalassiosira* sp. and *Chaetoceros* sp. (Crenn et al., 2018), fed to shrimp during these two stages. Similarly, Comamonadaceae family is observed in the postlarval stage of *L. vannamei* in association with *Artemia* (Tkavc et al., 2011), which has been given as feed diet at this stage. This bacterial group shows low abundance in juvenile shrimp after the diet has been changed to a commercial feed diet (Huang et al., 2016). Therefore, diets are one of the factors that can affect the bacterial composition of the microbiota in the growth development of shrimp.

In addition to animal host-related factors, the initial establishment of the host microbiome may be regulated by the rearing environment (Hasan and Yang, 2019; Scepanovic et al., 2019). Compared to terrestrial animals, aquatic organisms have a closer relationship with the microbiome from their aquatic environment (De Schryver and Vadstein, 2014). Therefore, surrounding environment bacteria significantly impact gut bacterial colonization in aquatic animals. In juvenile and adult shrimp, several studies have reported that gut microbiota is significantly affected by the microbial communities present in their aquatic environment, such as rearing water and pond sediments (Chen et al., 2017a; Cornejo-Granados et al., 2017; Rungrassamee et al., 2014). However, rearing water seems to affect the bacterial microbiota since the early life stages of the shrimp (Angthong et al., 2020; Pangastuti et al., 2010). For instance, bacterial genera *Ilumatobacter* and *Microcella* are shared between shrimp and their rearing water at early developmental stages of *P. monodon* (nauplius, zoea, mysis and postlarva) (Angthong et al., 2020). Bacterial communities of Pacific blue shrimp (*L. stylirostris*) do not differ significantly between eggs, nauplii and their rearing environment (Giraud et al., 2022), suggesting that in the early life stages of *L. stylirostris*, a vertical transfer from their broodstocks and rearing water to larvae through adhesion processes on the surface of these stages where no additional diets have been provided (Pangastuti et al., 2010). In larvae of *L. vannamei*, the similarity of bacterial communities between nauplii shrimp (pre-mouth opening) and rearing water bacteria is low and become highly similar at the zoea stage when they start feeding via their mouths (Wang et al., 2020b). This suggests that regulation of shrimp larval microbiota by microbial manipulation of rearing water, such as the application of probiotics, should begin at pre-mouth opening stage of shrimp larvae. Additionally, it has been shown that the salinity of the rearing environment is an important factor influencing the structure of the gut microbiota of shrimp (Chaiyapechara et al., 2022; Hou et al., 2017, 2020). Reducing water salinity may reduce the number of opportunistic *Vibrio* pathogens in the shrimp aquaculture (Chaiyapechara et al., 2022; Deris et al., 2022). The intestinal microbiota of *L. vannamei* postlarva is shifted following a reduction in rearing salinity, with an increase in potential pathogen species such as *V. parahaemolyticus* and *V. campbellii* and *V. owensii* (Bauer et al., 2021; Cao et al., 2020). On the other hand, innate immune responses could be directly affected by the changes in water salinity (Chaiyapechara et al., 2022), which may increase shrimp susceptibility to bacterial infection.

The composition of the microbiome can be affected by different farming practices. Some farming practices with poor water management, poor nutrition, and waste treatment can affect the composition of the environmental and gut microbiome, leading to dysbiosis of bacterial profiles and, thus higher risk of disease infections such as vibriosis (Callac et al., 2022; Holt et al., 2021). On the other hand, hatcheries typically employ biosecurity measures such as disinfection, quarantine, pathogen screening, and water filtration systems to remove waste to prevent the spread of disease in the system. Despite a highly-controlled management, reports of emerging diseases in hatcheries, such as acute hepatopancreatic necrosis disease (AHPND) in penaeid shrimp (Reyes et al., 2022) and hepatopancreatic microsporidiosis (Wan Sajiri et al., 2021), are accumulating and pose a high risk to larval development and survival. Therefore, balancing the gut microbiota of shrimp could be an important approach to increase disease resistance and improve survival. Diseases affecting the early developmental stages of shrimp are one of the main constraints to the sustainability of shrimp farming. Understanding the microbiota associated with early-life shrimp will help identify biomarkers that indicate the health status of host shrimp. For example, the bacterial genus *Meridianimaribacter* was significantly abundant in healthy *L. vannamei* larvae, which was reported as a potential probiotic candidate (Xue et al., 2016; Zheng et al., 2017), while *Nautella* was enriched in diseased shrimp (Zheng et al., 2017). Moreover, bacteria associated with AHPND infected *L. vannamei* larvae (mysis and postlarval stages) have been compared between groups with higher and lower survival (Reyes et al., 2022). They found high abundance of bacterial genera such as *Bacillus*, *Vibrio*, *Yangia*, *Roseobacter*, *Tenacibaculum*, *Bdellovibrio*, *Mameliella*, and *Cognatishimia* in the higher survival tank. In contrast, *Gilvibacter*, *Marinibacterium*, *Spongiimonas*, *Catenococcus,* and *Sneathiella* were found in lower abundant in the low survival group, suggesting that bacteria might be directly or indirectly affected by the presence of the pathogen.

3. Associations between early-life microbiota and host immune development

3.1. Shrimp immune system

Shrimp rely on the innate immune system as the first line of defense against pathogenic microorganisms, including cellular and humoral immune responses (Zhang et al., 2021). Cellular immune responses consist of pattern recognition proteins (PRPs) to recognize pathogen-associated molecular patterns (PAMPs) presented on the surface of pathogens and stimulate immune responses such as phagocytosis, apoptosis, encapsulation, and nodule formation (Jiravanichpaisal et al., 2006; Li and Xiang, 2013a). In contrast, humoral immune responses are activated by enzymes from factors in the hemolymph such as the blood clotting system, prophenoloxidase (proPO) system, and antimicrobial peptides (AMPs) (Söderhäll and Cerenius, 1998; Tassanakajon et al., 2018). The gastrointestinal tract of shrimp is the main route of entry for pathogens to colonize and develop into an infection in the crustacean (Duan et al., 2018; Soonthornchai et al., 2015). Shrimp build a physical barrier composed of chitin in their intestine called the peritrophic matrix (PM), which provides a protective barrier to guard against invading pathogens and can lead to stimulation of the immune system to maintain host homeostasis (Hegedus et al., 2009; Soonthornchai et al., 2015). Although the digestive system of shrimp is fully developed at the postlarval stage (Motoh, 1985; Ribeiro, 2010), their immune system is not as well developed as juvenile and adult shrimp (Angthong et al., 2023; Jiravanichpaisal et al., 2007; Soonthornchai et al., 2010). However, previous studies have shown that shrimp at early developmental stages (nauplius, zoea, mysis, and postlarva) also express primary immune responses, such as immune-related genes involved in PRPs, proPO system, immune deficiency (IMD) pathway, Toll pathway, and AMPs. For example, the transcript level of proPO increased with the larval developmental stage of *P. monodon* (Angthong et al., 2021; Jiravanichpaisal et al., 2007). The proPO system is one of the most important innate immune responses in shrimp. It is initially induced into an active form by the binding of PRPs to microbial membrane components, leading to the activation of the melanization process (Amparyup et al., 2013; Söderhäll and Cerenius, 1998). AMPs are also important for the host's innate immune system, exerting an antimicrobial effect on pathogenic bacteria (Silveira et al., 2018; Tassanakajon et al., 2018). In *P. monodon* and *L. vannamei* larvae, the members of AMP transcript families such as *crustin*, *anti-lipopolysaccharide factor* (*ALF*), *antiviral protein*, *peroxinectin,* and *penaeidin* have been observed to also increase with the developmental stage of shrimp (Angthong et al., 2021; Jiravanichpaisal et al., 2007; Tassanakajon et al., 2010). Several isoforms of the *Toll-like receptors* (*TLRs*) transcript have been reported to be constitutively expressed during larval development of *P. monodon* (Angthong et al., 2021), which have been described as part of the shrimp defense mechanism component in the Toll pathway against pathogenic microorganisms (Deepika et al., 2020; Kawasaki and Kawai, 2014; Mohd Ghani and Bhassu, 2019). Toll receptors recognize presence of a pathogen component, leading to activation of the Toll signaling cascade and AMPs synthesis (Kawasaki and Kawai, 2014; Tassanakajon et al., 2013). *TLRs* have been reported to be expressed by larval and adult shrimp under both non-pathogenic and pathogenic conditions (Deepika et al., 2020; Habib et al., 2021; Sreedharan et al., 2018; Yang et al., 2008), suggesting that they serve as part of the primary innate immune response in shrimp. In addition, other components of the Toll pathway such as *Spätzle* and *Tube, Pelle, and tumor necrosis factor receptor-associated factor 6* (*TRAF6*), also showed constitutive expression in early developmental stages of *P. monodon* (Angthong et al., 2021). The increased expression of *TRAF6* has been reported in *P. monodon* postlarvae and adults during *V. harveyi* challenge (Deepika et al., 2020; Sreedharan et al., 2018). The Toll signaling pathway may provide for a basic immune system established in shrimp at early larval stages.

3.2. Evidence of crosstalk between shrimp immune system and microbiota

Understanding the complex interplay between the immune system and the microbiota is important to improve disease management, enhance immune responses, develop probiotics and prebiotics, and improve environmental stress resistance in shrimp aquaculture (Holt et al., 2021). Like many other animal hosts, host-gut microbial interactions in shrimp are also complex and highly dynamics (Chen et al., 2017b). To date, understanding of the association between gut microbiota and immunity in penaeid shrimp remains relatively limited and requires tools such as the gnobiotics model to enable an in-depth investigation to determine a causal relationship. Such a relationship has been mostly derived from the invertebrate model organisms such as *Drosophila*, due to their advantages of being to be reared in sterile or germ-free environments (Douglas, 2018). For example, when comparing *Drosophila melanogaster* rearing under conventional environment to those reared under a germ-free environment gene expression pattern is significantly altered in the guts of sterile *D. melanogaster* (Broderick et al., 2014), indicating that the gut microbiome can influence on host gene expression and the absence of microbiome results in the lack of stimulation (Bost et al., 2018; Elya et al., 2016). For instance, the presence of gut microbes has been shown to trigger the expression of genes involved in cellular signaling, immune and stresses responses such as those genes involved in the insulin-like signaling (IIS) and target of rapamycin (TOR) signaling pathways as well as transcription factors such as the IMD factor *Relish* (Dobson et al., 2016). In *Drosophila* exposed to beneficial bacteria, such as *Lactiplantibacillus plantarum*, expression of the peptidoglycan-recognition protein (*PGRP-SC1*) is reduced, which is linked to improved growth performance (Gallo et al., 2022). Nevertheless, there is growing evidence of crosstalk between the shrimp immune system and microbiota, indicating the interactive roles of the microbiome in regulating the immune response of shrimp (Angthong et al., 2023; Duan et al., 2018; Zhang and Sun, 2022). The gut microbiota of shrimp can influence the expression of immune-related genes, such as those involved in antimicrobial peptide production and phagocytosis (Chiu et al., 2007; Kongnum and Hongpattarakere, 2012). The microbiota can also modulate the shrimp's immune response to pathogens. For example, a study on Pacific white shrimp found that certain bacteria groups in the gut were associated with increased resistance to white spot syndrome virus (WSSV) (Niu et al., 2019; Pilotto et al., 2018). On the other hand, the host intestinal immune system plays a critical role in maintaining the balance between the gut microbiota and the host. For instance, RNAi technology has been employed in the black tiger shrimp to silence STAT, a part of the JAK/STAT pathway, which regulates immune responses in invertebrates (Li and Xiang, 2013b). Shrimp gut microbial diversity is significantly altered when STAT is silenced, suggesting that STAT could also play a role in maintaining microbial homeostasis in the shrimp gut environment (Jatuyosporn et al., 2023). In Kuruma shrimp (*Marsupenaeus japonicus*), small antibacterial effectors, such as lysozymes, lectins, and antimicrobial peptides, have been shown to be important in maintaining intestinal immunity (Liu et al., 2022). These proteins work together to synergistically regulate peptidoglycan-derived intestinal immunity. In particular, LysC, a C-type lysozyme, has been identified as a key factor in the interaction between the intestine and bacteria. LysC can regulate intestinal bacteria by, for example, cleaving the peptidoglycan of *Photobacterium damselae,* which indirectly leads to muropeptides. Muropeptides then induce the expression of *anti-lipopolysaccharide factor B1* (*AlfB1*) and maintain intestinal homeostasis. By using RNAi as a tool to knock down the expression of various C-type lectin genes in Kuruma shrimp, some groups of C-type lectins, such as *Ctl24,* can affect the abundance of intestinal bacteria. Therefore, this study demonstrates the interactions between the host and gut microbiota in Kuruma shrimp. C-type lectins play a role in controlling intestinal homeostasis, and peptidoglycans are important factors in this relationship (Liu et al., 2022). Thus, the microbiota and the host immune system have a complex and bidirectional relationship. The microbiota can influence the host's immune system, and the host's immune system can also shape the composition and function of the microbiota. More studies in this area will lead to a better understanding of the crosstalk between the shrimp immune system and microbiota, which has the potential to revolutionize shrimp aquaculture.

4. Challenges and pitfalls in shrimp microbiome research

Microbiome in shrimp has gathered much attention in the past decade with its promising potentials to help with various aspects of aquaculture production. While the application of probiotic, prebiotic, and synbiotic to shape the animal's microbiome to stimulate the immune response and reduce disease occurrence might have become common knowledge to aquaculturists, microbiome may have far reaching potentials such as tools for predicting disease outbreaks, monitoring water quality or even part of the calculation for biosecurity and environmental friendliness (Bentzon-Tilia et al., 2016; Gutiérrez-Pérez et al., 2022). Similar to microbiome research in other fields, there are several challenges and pitfalls that needs to be addressed to ensure the accuracy and reliability of the findings. This section presents a summary of these important issues and provide the readers with some guides to assess the results.

With over 50,000 research articles published, the number of research articles on gut microbiome have grown over 30 times in the two decades between 2000 and 2020 including many in highly respected journals (Li et al., 2020). Several review and commentary papers raising various issues on the microbiome research had been published over the years and common themes have emerged. In 2014, five crucial questions were raised to evaluate and conduct microbiome research, including the experimental design, causation vs. correlation, the need to understand the mechanism, whether the experiment reflected reality, and whether the microbiome was the only cause of the observed results (Hanage, 2014). These questions are still valid today to assess research articles on the microbiome. In the commentary paper, it emphasizes the importance of microbiome research and how the research community needs to consider various criticisms of result interpretation (Li et al., 2020). For example, with more research on understanding the mechanism, the gut microbiome's influence on important human health and disease conditions such as obesity and the host's metabolism has been established, making the jump from correlation to causation. With a better understanding, the use of gut microbiome as a diagnosis and monitoring tool could also become available. Still, the mechanism of the microbiome's influence on many aspects of human health remains to be elucidated. In turn, our understanding of the roles, including the mechanism of the microbiome on the health of shrimp was even more limited. Other additional issues mentioned include standardization, repeatability, the long-term efficacy of microbiome intervention, and over-interpretation of results (Li et al., 2020).

4.1. Experimental design, methodological variation, and sampling

With its interdisciplinary nature and multiple confounding factors, researchers must use extra caution in designing a microbiome experiment. Various issues relating to the design of the experiment and the reproducibility of the results have been a cause for concern in microbiome research (Schloss, 2018). For the typical analyses of 16S rRNA gene sequencing, some important considerations included DNA extraction method (choice of extraction kits, pre-treatment), choice of primers, and sequencing platform. Each method has advantages and limitations, and researchers must carefully select the most appropriate method for their study. Additionally, different laboratories may use different protocols, which can lead to methodological variation and affect the comparability of results. In a review paper on the roles of shrimp gut microbiome on health and disease (Holt et al., 2021), summarized findings from previous research on shrimp gut microbiome. They showed the variety of the 16S variable region and sequencing platform chosen by researchers. The most often used region for shrimp gut microbiome sequencing was the V3–V4 variable region, which was sufficient to characterize the gut microbiome of shrimp (García-López et al., 2020). However, identifying the microbial taxa present in a shrimp microbiome can be challenging due to the high degree of diversity and the presence of rare taxa.

The use of standard controls in multiple steps of the experiment can also be employed to improve the reproducibility and validity of microbiome results (Hornung et al., 2019). examined the use of these standards in a review paper and suggested using both negative and positive controls during sampling, extraction, and sequencing. Including DNA of mock communities as spike-in standards (Tourlousse et al., 2016) at various steps can help with quality control of the experiment, and the option to add mock communities DNA during sequencing is now available for commercial service.

Another important consideration is that of sample size and power analysis. Shrimp microbiomes can vary depending on age, diet, and environment. Therefore, it is crucial to ensure enough sampling to capture the full range of microbial diversity and account for the natural variation expected in the samples. The number of samples required to detect a meaningful difference in the results should be calculated and planned before the experiment. Examples of power calculation and sample size estimated can be found for clinical microbiome analysis (Casals-Pascual et al., 2020; Kelly et al., 2015; Kers and Saccenti, 2022). While most of the papers suggested using both alpha and beta diversity with various diversity indices to calculate samples, beta diversity using the Bray-Curtis index was shown to be the most sensitive and yield the lowest required sample size (Kers and Saccenti, 2022). For shrimp microbiome research, the suitable sample size required are seldom estimated. It is difficult to generalize for all types of experiments how many replicates and shrimp per sampling would be sufficient. In many cases, researchers may be limited by the size and the number of tanks or aquaria available for an experiment. However, it's probably best to plan the experiment with the number of replicates in mind for sufficient statistical analyses.

Due to the size of the shrimp during early developmental stages, pooling of individual animals was required to obtain sufficient high-quality DNA for analysis. The range of individuals in a pool for early developmental white shrimp reported in the literature is 200–500 for zoea, 80–120 for mysis, and 15–50 for postlarvae (Angthong et al., 2020; Zheng et al., 2017). Researchers must keep in mind the effect of the pooling of samples and whether it could be appropriate for the design of their studies. High heterogeneity among individual shrimp can be observed even when they were reared in a similar environment (i.e., within the same replicate tank). Pooling of samples can neutralize this variation, which can be suitable for studies examining the treatment effects on a population (Ray et al., 2019). At the same time, the pooling of samples may not be appropriate for examining effects on individuals in a population or in a diversity survey of a wild population (Adair et al., 2018; Rodríguez-Ruano et al., 2020; Wang and Staubach, 2018). It is also important to remember that the number of individuals in a pool does not affect the statistical power and that sufficient numbers of true replicates are always required in the experimental design.

4.2. Data analysis, statistical analysis, and functional inference

Microbiome data analysis can be complex and requires specialized expertise. Different statistical methods need to be employed as the research in microbiome shifted from descriptive study to hypothesis testing (Buttigieg and Ramette, 2014; Xia and Sun, 2017). With the increasing multidisciplinary approach for microbiome study, advanced statistical analyses such as multi-omics network analysis (Jiang et al., 2019) or even machine learning (Jiang et al., 2022). At various steps of analysis, such as bioinformatic and subsequent statistical analysis, the choice of method used can influence the interpretation of the results. An example of how statistical analysis can influence the results can be found in (Khomich et al., 2021). While it might be difficult to assess which methods are best, it is prudent to devise the data analysis and statistical analysis scheme in advance and stick to the plan. Researchers must avoid common pitfalls, such as overfitting, false discovery rates, and multiple testing corrections. In addition, searching for the "right" statistical analysis to provide significant results (p-hacking) must be avoided. In addition to use appropriate statistical methods, the interpretation of data should also be done in biologically meaningful way.

Understanding the functional roles of microbial communities in the shrimp microbiome requires additional analysis beyond taxonomic identification. Functional prediction based on 16S rRNA gene inference such as PICRUSt and Tax4Fun was frequently used to obtain functional profiles of the microbial communities (Aβhauer et al., 2015; Douglas et al., 2018). However, the accuracy and reliability of these functional predictions depend on the database of both the microbiome and the metagenome of the bacteria. While these prediction tools may have good performance in human microbiome, the prediction in the non-human dataset and environmental samples are not as reliable (Sun et al., 2020). For these types of non-human samples, researchers must still rely on techniques such as metagenomics or metatranscriptomics to infer the functional potential of the microbiome and its impact on shrimp health.

4.3. Challenges for the association between microbiome and immunity study

Further advancement in our understanding of the involvement of microbiome and immune development in shrimp relies on a good experimental design and integration of different multi-omics analyses. Still, based on the current literature at the time of this review, gnotobiotic models to test the complex interaction between the microbiome and the immune system in shrimp has not been developed.

During the results interpretation phase of the study of the microbiome-immune response interaction, researchers would often end up with a chicken-and-egg dilemma at some point. Whether the change in the microbiome after an experimental manipulation causes the change in the immune response or vice versa is always a point of major discussion. When the factors being studied and manipulated in the experiment were not solely impacting either the microbiome or the immune system but can affect both (such as salinity and temperature), it can be particularly difficult to come to a meaningful conclusion. Using gnotobiotic animal can allow for the comparison of the microbe-immune response interaction with and without the bacteria. Gnotobiotic crustaceans, including *Daphnia magna* and *Artemia franciscana,* have been used for experiments (Baruah et al., 2015; Marques et al., 2006; Zhang et al., 2020). These crustaceans were made anoxic using varieties of methods such as repeated washing, antibiotic treatment or disinfection, and their germ-free status was confirmed using plate count or direct count of microorganisms. Gnotobiotic shrimp could most likely be created during the early developmental stage, but the main challenges would be establishing the germ-free live feed for this early development shrimp. The establishment of gnotobiotic shrimp would allow for more detailed and more specific study of the interaction between microbiome and immune responses.

Tools and techniques from related field of studies can be employed to support the study of microbe-immune interaction. Histological analysis, fluorescence in situ hybridization (FISH), physiological measurement, and others in combination with multi-omics technology can be immensely helpful in our understanding of the interaction. Metabolome analysis could also be considered to monitor the metabolite roles between the microbiome and the host immunity. These additional techniques could add new perspective to the understanding of the interaction.

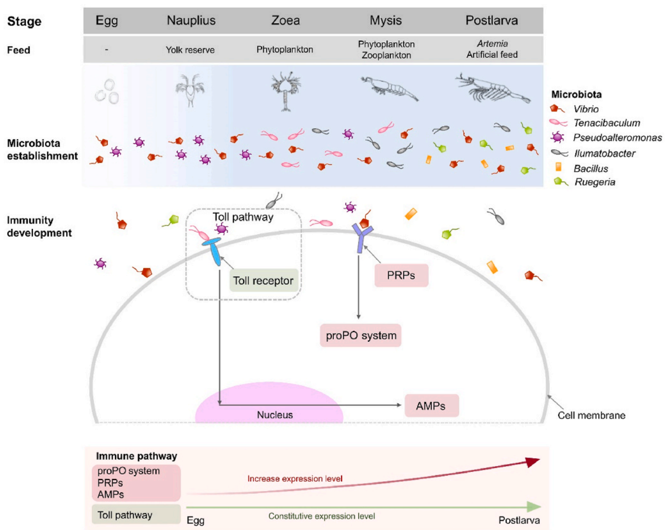

Fig. 2. Gut microbiota and immune system establishment in early penaeid shrimp development. Pattern recognition proteins (PRPs), prophenoloxidase system (proPO system), antimicrobial peptides (AMPs), and Toll pathway were involved in the early life stages of penaeid shrimp.

5. Perspectives and concluding remarks

Microbiome research has gathered a lot of momentum in the past decade. At present, research in microbiomes has come a long way from the simple descriptive analysis of the bacterial community to more hypothesis testing. This review summarizes the current understanding of the establishment of gut microbiota in the early life of penaeid shrimp (Fig. 2). The early developmental stages are a critical period, as the composition of the microbiota is shaped during this time. The microbiota of penaeid shrimp eggs is initially sterile, but it begins to colonize within hours or days of metamorphosis to nauplius, zoea, mysis, and postlarvae. The initial colonization is influenced by several factors, including the hatchery environment and feed diet. The dominant microbiota includes species that are commonly found associated with marine animals and environments, such as *Vibrio*, *Tenacibaculum*, *Pseudoalteromonas*, *Illumatobacter*, *Bacillus*, and *Ruegeria*. In parallel, a transcriptomic analysis of one penaeid shrimp species, *P. monodon*, shows that immune-related genes are constitutively expressed in the nauplius stage. Notably, transcripts of *Spätzle* and *Toll receptors* are expressed in this stage, suggesting that the Toll pathway serves as a foundational immune system during early larval stages. As the larvae develop further, genes encoding the prophenoloxidase system (proPO system), antimicrobial peptides (AMPs), and pattern-recognition proteins (PRPs) are modulated, indicating the establishment of immune defense mechanisms in later larval stages. However, the direct interactions or causal relationship between the establishment of the gut microbiota and the host immune system have not yet been determined.

With increasing knowledge of the roles of microorganisms on the host animals and the cost of next-generation sequencing is no longer prohibitive, this provides indispensable tools to advance this study. Nevertheless, researchers must apply the scientific rigor necessary for microbiome studies as with all other types of research. It is crucial to carefully plan for microbiome analysis and not to treat it as an add-on to increase the novelty or appeal of one's research. Care should be exercised in the experimental design, sample size, methodology, data and statistical analysis, and interpretation. Additional analyses to validate the microbiome results supporting the research question should be considered whenever possible. With the new knowledge of the interaction between the microbiome and immune responses, more targeted, more effective, and more reliable probiotics can be developed and assessed. Alternative methods to shape and promote the microbiome balance can be devised. For economically important aquaculture species, the knowledge of the microbe-immune interaction and its potential application would provide a novel solution for animal health and disease management.

Declaration of competing interest

The authors declare that they have no competing financial interests.

Data availability

No data was used for the research described in the article.

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